

```
In [1]: import os

print(os.getcwd())
```

C:\Users\user

```
In [2]: import pandas as pd

file_path = r"C:\Users\user\Downloads\heart_attack_dataset (1).csv"

df = pd.read_csv(file_path)

print(df.head())
```

	patient_id	age	gender	chest_pain_type	resting_bp	cholesterol	\
0	1	59	Male	Asymptomatic	105.0	154.0	
1	2	50	Male	Non-anginal Pain	102.0	180.0	
2	3	61	Male	Atypical Angina	117.0	213.0	
3	4	73	Female	Atypical Angina	121.0	208.0	
4	5	49	Male	Non-anginal Pain	106.0	157.0	

	fasting_blood_sugar	resting_ecg	max_heart_rate	exercise_angina	\
0	0.0	ST-T Abnormality	186.0	1	
1	0.0	Normal	183.0	0	
2	1.0	Normal	161.0	0	
3	1.0	Normal	150.0	0	
4	0.0	Normal	185.0	0	

	...	num_major_vessels	thalassemia	bmi	smoking_status	\
0	...	1	Fixed Defect	23.5	Former	
1	...	0	Reversible Defect	19.3	NaN	
2	...	1	Fixed Defect	30.5	Never	
3	...	0	Normal	28.6	Former	
4	...	0	Reversible Defect	21.4	Former	

	alcohol_consumption	physical_activity	family_history	diabetes	\
0	Heavy	Moderate	1	1	
1	Moderate	High	0	0	
2	Moderate	Low	1	1	
3	Moderate	Moderate	0	0	
4	Moderate	Moderate	0	0	

	stress_level	heart_attack_risk
0	6.0	0
1	8.0	0
2	2.0	1
3	4.0	0
4	NaN	0

[5 rows x 22 columns]

```
In [3]: print(df["smoking_status"].isnull().sum())
```

505

```
In [4]: print(df["smoking_status"].unique())
```

```
<StringArray>
['Former', nan, 'Never', 'Current']
Length: 4, dtype: str
```

```
In [5]: mode_value = df["smoking_status"].mode()[0]
```

```
print("Mode:", mode_value)
```

```
Mode: Never
```

```
In [6]: df["smoking_status"] = df["smoking_status"].fillna(mode_value)
```

```
# Verify
print(df["smoking_status"].isnull().sum())
```

```
0
```

```
In [7]: print(pd.crosstab(df["smoking_status"], df["heart_attack_risk"]))
```

heart_attack_risk	0	1
smoking_status		
Current	726	811
Former	970	740
Never	2364	1389

```
In [8]: print(pd.crosstab(df["smoking_status"], df["heart_attack_risk"], normalize="index"))
```

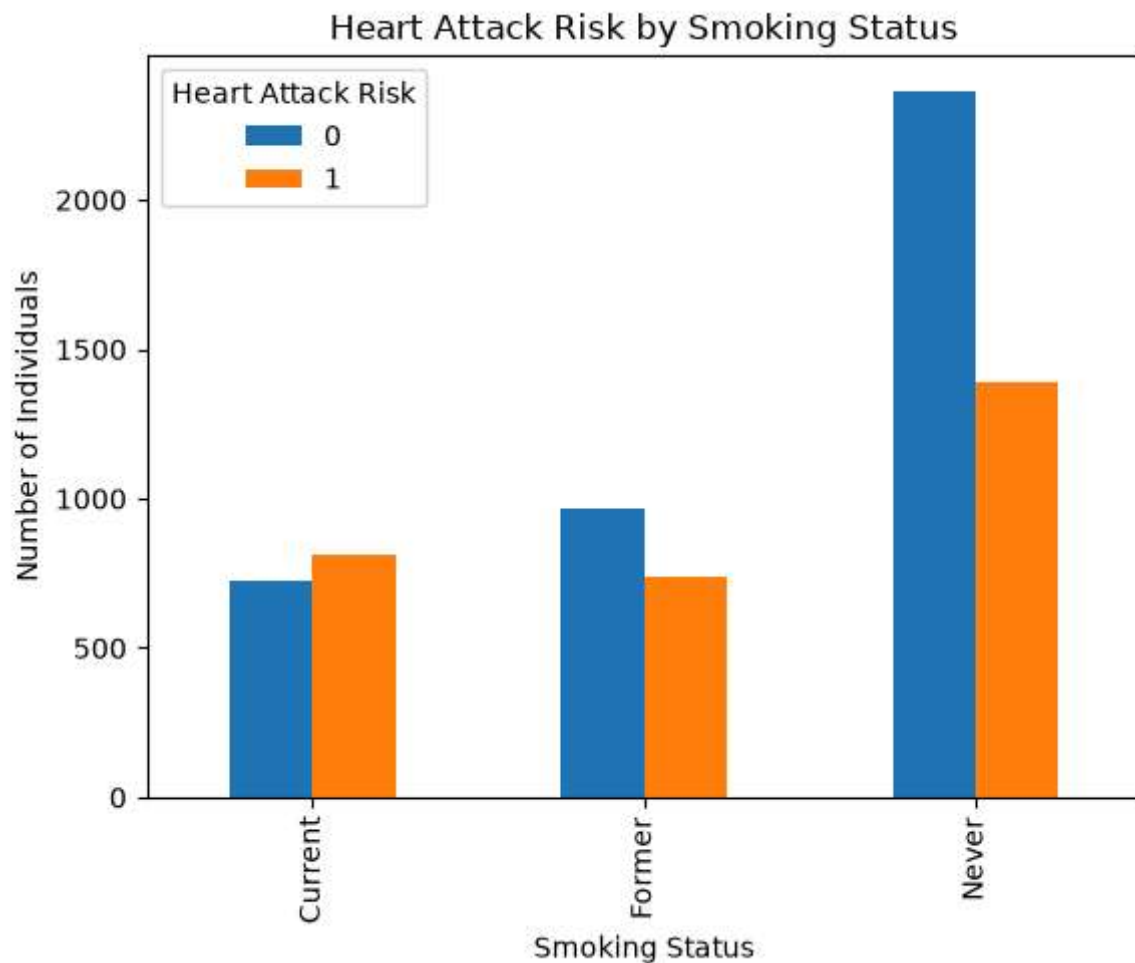
heart_attack_risk	0	1
smoking_status		
Current	47.234873	52.765127
Former	56.725146	43.274854
Never	62.989608	37.010392

```
In [ ]: it shows heart attack risk increases from 37% never smoking status to 53% current s
```

```
In [9]: import matplotlib.pyplot as plt
import pandas as pd
```

```
pd.crosstab(df["smoking_status"], df["heart_attack_risk"]).plot(kind="bar")
```

```
plt.title("Heart Attack Risk by Smoking Status")
plt.xlabel("Smoking Status")
plt.ylabel("Number of Individuals")
plt.legend(title="Heart Attack Risk")
plt.show()
```



```
In [10]: print(df["alcohol_consumption"].isnull().sum())
```

405

```
In [11]: print(df["alcohol_consumption"].unique())
```

```
<StringArray>
['Heavy', 'Moderate', nan, 'Non-drinker']
Length: 4, dtype: str
```

```
In [12]: mode_value = df["alcohol_consumption"].mode()[0]
```

```
print("Mode:", mode_value)
```

Mode: Non-drinker

```
In [13]: df["alcohol_consumption"] = df["alcohol_consumption"].fillna(mode_value)
```

```
# Verify
```

```
print(df["alcohol_consumption"].isnull().sum())
```

0

```
In [14]: import pandas as pd
```

```
risk_table = pd.crosstab(df["alcohol_consumption"], df["heart_attack_risk"])
```

```
print(risk_table)
```

heart_attack_risk	0	1
alcohol_consumption		
Heavy	538	461
Moderate	1502	1128
Non-drinker	2020	1351

```
In [15]: risk_percent = pd.crosstab(
    df["alcohol_consumption"],
    df["heart_attack_risk"],
    normalize="index"
) * 100

print(risk_percent.round(2))
```

heart_attack_risk	0	1
alcohol_consumption		
Heavy	53.85	46.15
Moderate	57.11	42.89
Non-drinker	59.92	40.08

In []: it shows alcohol drinking heavy drinking status is 46% as compared to non drinker s

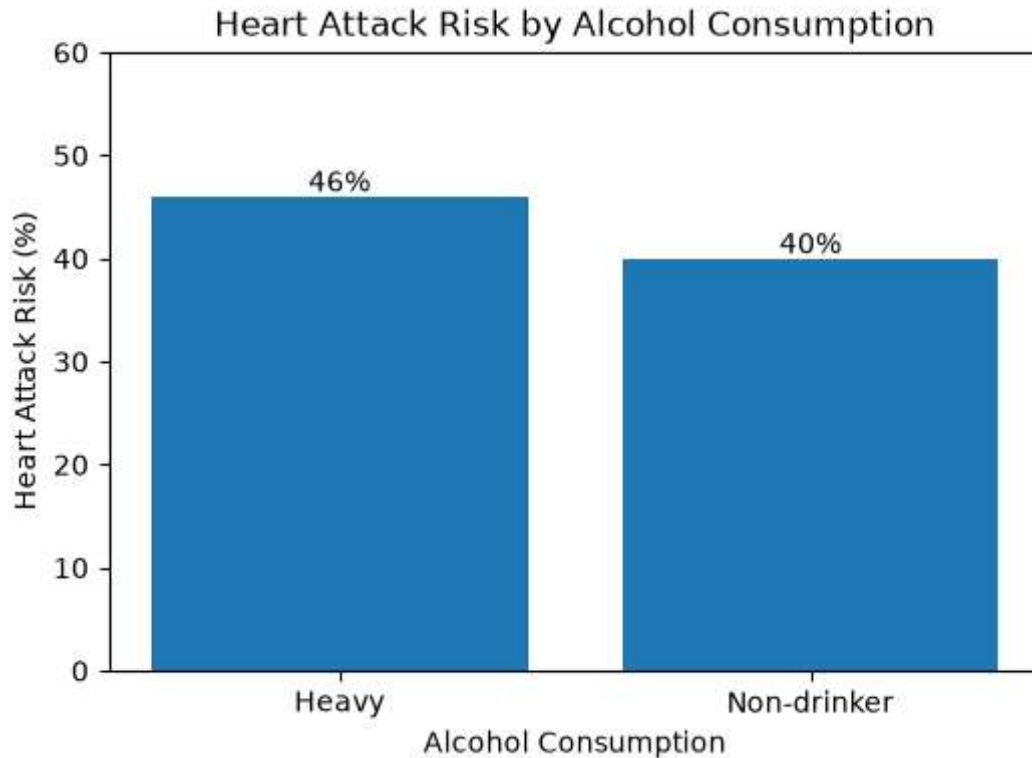
```
In [16]: categories = ["Heavy", "Non-drinker"]
percentages = [46, 40]

# Create bar chart
plt.figure(figsize=(6,4))
plt.bar(categories, percentages)

# Labels and title
plt.title("Heart Attack Risk by Alcohol Consumption")
plt.xlabel("Alcohol Consumption")
plt.ylabel("Heart Attack Risk (%)")

# Show percentage on top of bars
for i, value in enumerate(percentages):
    plt.text(i, value + 0.5, f"{value}%", ha="center")

plt.ylim(0, 60)
plt.show()
```



```
In [17]: import pandas as pd

risk_table = pd.crosstab(df["physical_activity"], df["heart_attack_risk"])

print(risk_table)
```

heart_attack_risk	0	1
physical_activity		
High	793	455
Low	1366	1259
Moderate	1680	1101

```
In [18]: risk_percent = pd.crosstab(
    df["physical_activity"],
    df["heart_attack_risk"],
    normalize="index"
) * 100

print(risk_percent.round(2))
```

heart_attack_risk	0	1
physical_activity		
High	63.54	36.46
Low	52.04	47.96
Moderate	60.41	39.59

```
In [ ]: Heart attack risk with high physical activity is 36% while it increases to 48%with
```

```
In [19]: import matplotlib.pyplot as plt

# Data
activity = ["High", "Low"]
```

```

risk = [36, 48]

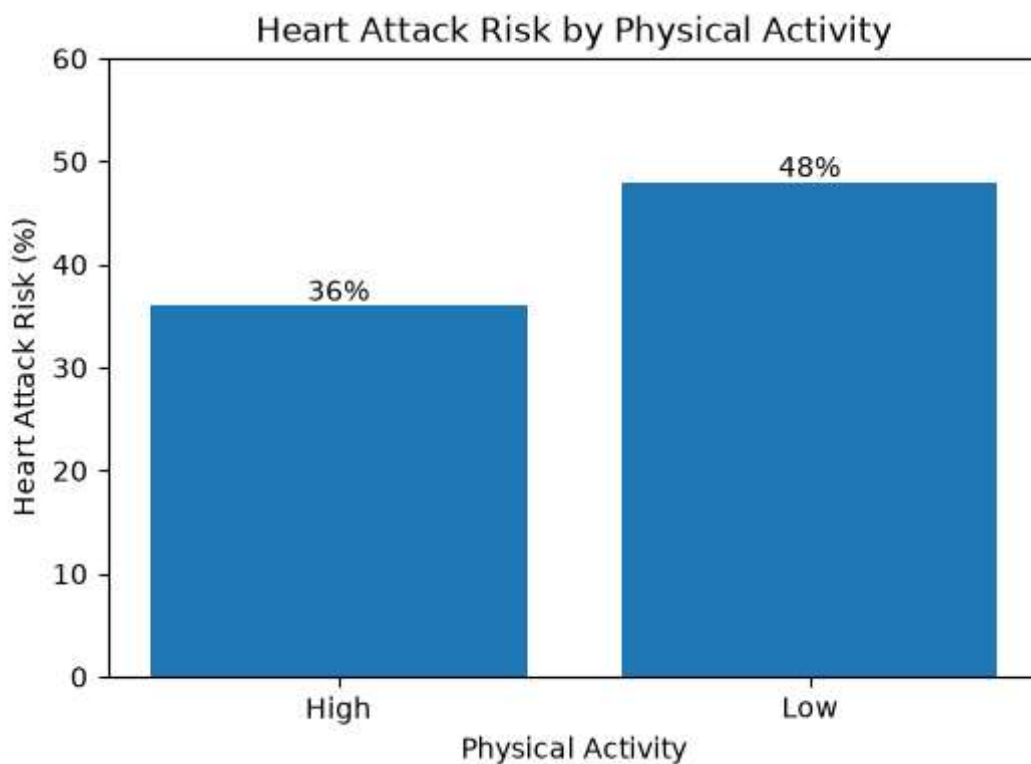
# Create bar chart
plt.figure(figsize=(6,4))
plt.bar(activity, risk)

# Add Labels
plt.title("Heart Attack Risk by Physical Activity")
plt.xlabel("Physical Activity")
plt.ylabel("Heart Attack Risk (%)")

# Display values on bars
for i, value in enumerate(risk):
    plt.text(i, value + 0.5, f"{value}%", ha="center")

plt.ylim(0, 60)
plt.show()

```



```

In [21]: age_risk = df.groupby("heart_attack_risk")["age"].mean()

print(age_risk)

```

```

heart_attack_risk
0    47.826355
1    57.572789
Name: age, dtype: float64

```

```

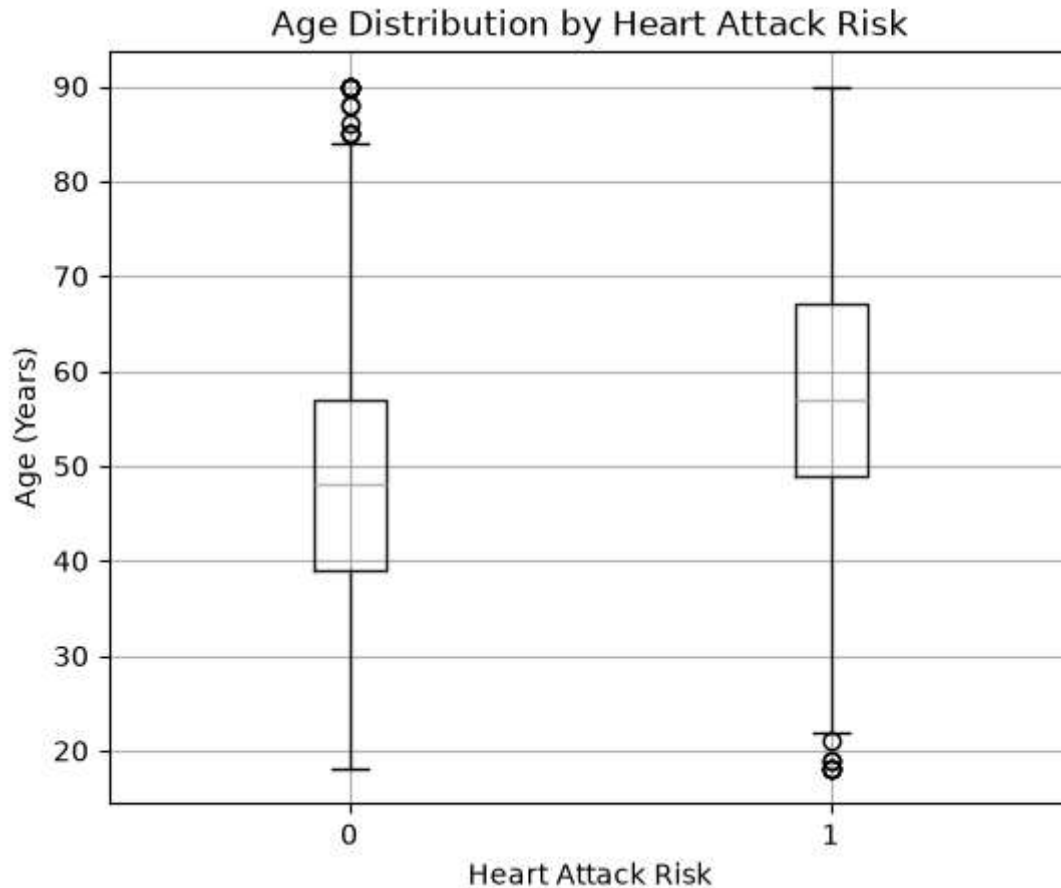
In [22]: import matplotlib.pyplot as plt

df.boxplot(column="age", by="heart_attack_risk", figsize=(6,5))

plt.title("Age Distribution by Heart Attack Risk")

```

```
plt.suptitle("") # Removes the default title
plt.xlabel("Heart Attack Risk")
plt.ylabel("Age (Years)")
plt.show()
```

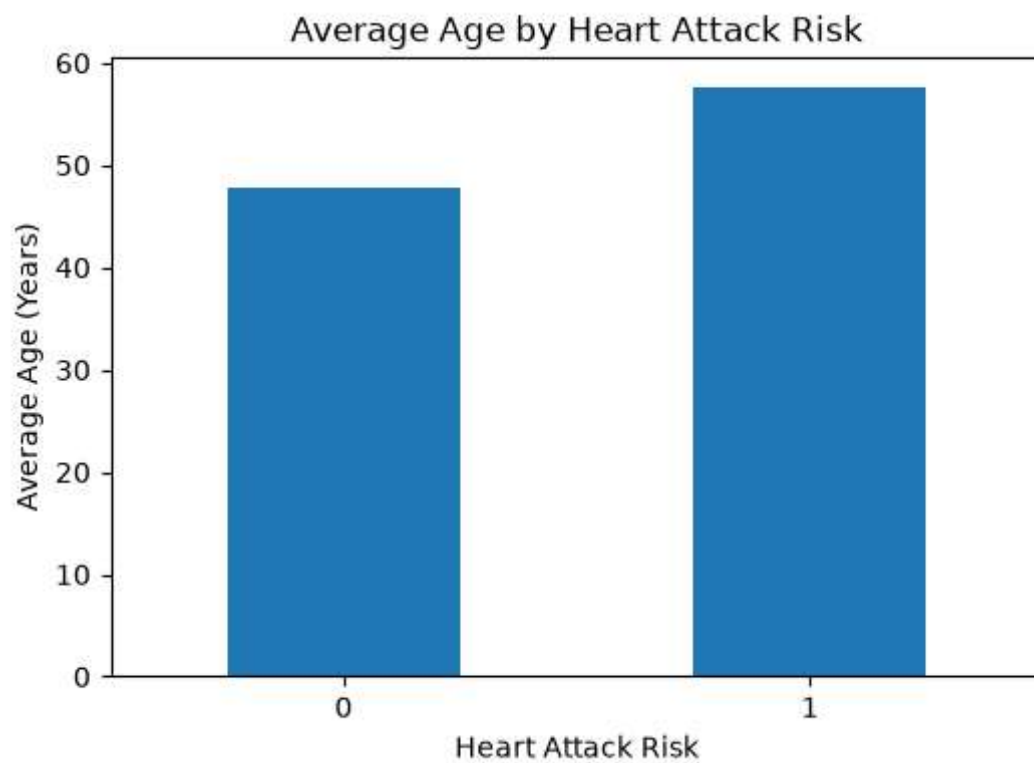


```
In [23]: import matplotlib.pyplot as plt

# Calculate mean age
age_risk = df.groupby("heart_attack_risk")["age"].mean()

# Plot
age_risk.plot(kind="bar", figsize=(6,4))

plt.title("Average Age by Heart Attack Risk")
plt.xlabel("Heart Attack Risk")
plt.ylabel("Average Age (Years)")
plt.xticks(rotation=0)
plt.show()
```



In []: near 60 years of age there is greater chance of heartattack.